Data management plan (DMP)

NO2 Air Quality Classification across Austrian Monitoring Stations

NO2-AQ-AT

Version	Effective date	Description of document/changes
1.0	30/05/2026	Final version of the DMP

FWF Data Management Plan (DMP)

I General Information	
I.1 Administrative information	**PI:** **FWF project number:** **Internal Project ID:** **DMP version:** v1.0 (Final DMP), 30/05/2026 **Contributors:** Sina Sadeghi, sina.sadeghi@student.tuwien.ac.at, TU Wien, ROR: ror.org/04d836q62, Researcher Muhammed Tayyab, e12510765@student.tuwien.ac.at, TU Wien, ROR: ror.org/04d836q62, Researcher Somayeh Zeraati, e12353396@student.tuwien.ac.at, TU Wien, ROR: ror.org/04d836q62, Researcher Glent Rembeci, e12332209@student.tuwien.ac.at, TU Wien, ROR: ror.org/04d836q62, Researcher
I.2 Data management responsibilities and resources	**Person responsible for data management and DMP:** Muhammed Tayyab, e12510765@student.tuwien.ac.at, TU Wien, ROR: ror.org/04d836q62 **Co-ordination of data management responsibilities across partners:** Muhammed Tayyab, e12510765@student.tuwien.ac.at, TU Wien, ROR: ror.org/04d836q62 **Resources costed in for RDM:** There are no costs dedicated to data management and ensuring that data will be FAIR.
II Data Characteristics	
II.1 Data description and collection or re-use of existing data	**Produced datasets:**

dataset ID	title	type	format	estimated volume	contains sensitive data	description
P1	NO2 Air Quality Classifier and Evaluation Outputs	Structured text, Images, Scientific and	joblib, json, csv, png	100 - 1000 MB	no	Trained Random Forest binary classifier predicting elevated NO2 concentrations (>= 40 µg/m³)

		statistical data formats				across Austrian monitoring stations. Includes the trained model artefact (joblib), test set predictions (csv), evaluation metrics (json), and confusion matrix visualisation (png). Accuracy 0.8005, precision 0.2107, recall 0.7251, F1 0.3266. Deposited in the TU Wien Research Data Repository.
P2	NO2 Air Quality DBRepo Database	Databases	SQL, CSV	100 - 1000 MB	no	Normalised relational database (3NF) of EEA NO2 measurements stored in the TU Wien DBRepo test instance. Contains 8 tables including measurements (140,160 rows), sampling points, pollutants, measurement units, aggregation types, validity flags, verification flags, and observation logs. All columns are annotated with semantic ontology concepts from SOSA/SSN and unit of measurement URIs from QUDT and the SI Digital Framework.

Reused datasets:

dataset ID	title	source	rights (e.g. license)	contains sensitive data	description
R1	EEA NO2 Air Quality Measurements Austria 2024	https://www.eea.europa.eu/en/datahub/datahubitem-view/778ef9f5-6293-4846-badd-56a29c70880d	CC BY 4.0	no	…

Methods and software used for data generation and reuse

The project reuses existing EEA NO2 air quality measurements downloaded from the European Environment Agency data hub as Parquet files. The data was normalised into a relational database schema (3NF) in DBRepo using Python and the DBRepo REST API. New data is generated through a Random Forest classification experiment implemented in Python using scikit-learn. The experiment produces a trained model, predictions, evaluation metrics, and visualisations. All data processing, model training, and evaluation scripts are available in the project GitHub repository.

III Documentation and Data Quality	
III.1 Metadata and documentation	**Data organisation, metadata, and documentation:** The input data is structured as 16 station-specific Parquet files in data/raw/ following EEA source file naming conventions. The normalised database in DBRepo uses a 3NF relational schema with 8 tables. All code and data files are version-controlled in a public GitHub repository using Git. Releases are tagged using semantic versioning (v1.0, v2.0) and automatically archived on Zenodo with a DOI per release. The DBRepo instance provides built-in data versioning with timestamped query support. The experiment is described using five complementary metadata standards: RO-Crate (experiment package), CodeMeta 2.0 (software), FAIR4ML (ML model), Croissant (input dataset), and a Model Card (human-readable model description). All dataset attributes are mapped to semantic ontology concepts using SOSA/SSN and domain ontologies. Numeric attributes are mapped to unit of measurement concepts from the SI Digital Framework and QUDT. Persistent identifiers (DOIs) are assigned via Zenodo, TUWRD, and DBRepo. This will help others to identify, discover and reuse our data. Additionally, we will provide common metadata such as title, description, or keywords when publishing data in open access repositories. In such a case, we will follow the default template provided by the repository, such as Data Cite Metadata or Dublin Core. As far as possible, we will use controlled vocabularies for our data to allow inter-disciplinary interoperability and machine-actionability. A comprehensive README.md provides step-by-step reproduction instructions for both local and DBRepo-based execution. All scripts are documented with inline comments. The FAIR4ML metadata file documents the algorithm, hyperparameters, evaluation metrics, intended use, and limitations. The Model Card provides human-readable documentation of the model. An RO-Crate metadata file describes all relationships between inputs, code, and outputs. Validation outputs for the RO-Crate are stored in docs/validation/.
III.2 Data quality control	**Data quality control**: The following data quality checks will be done: standardised data capture, representation with controlled vocabularies and Automated data validation through DBRepo schema constraints and foreign key relationships. EEA validity and verification flags are preserved as separate lookup tables to document data quality at source..
IV Data Storage, Sharing, and Long-Term Preservation	
IV.1 Data storage and backup during the research process	**Storage and backup facilities:** For the duration of the project, storage and backup of data will be ensured by Muhammed Tayyab (acting as the person responsible for data management and DMP) in cooperation with the system operator. The data will not be stored on the servers of TU Wien. R1 (EEA NO2 Air Quality Measurements Austria 2024) will be stored on GitHub. External storage will be used because github and zenodo are used alongside institutional storage (dbrepo and tuwrd) for version control, collaboration, and software citation purposes. github provides distributed version control and zenodo provides automatic doi minting integrated with github releases. all research data and model outputs are additionally preserved in tu wien institutional repositories (dbrepo and tuwrd). R1 (EEA NO2 Air Quality Measurements Austria 2024) will be stored on Zenodo. External storage will be used because github and zenodo are used

	alongside institutional storage (dbrepo and tuwrd) for version control, collaboration, and software citation purposes. github provides distributed version control and zenodo provides automatic doi minting integrated with github releases. all research data and model outputs are additionally preserved in tu wien institutional repositories (dbrepo and tuwrd). P1 (NO2 Air Quality Classifier and Evaluation Outputs) will be stored on TU Wien Research Data Repository. External storage will be used because github and zenodo are used alongside institutional storage (dbrepo and tuwrd) for version control, collaboration, and software citation purposes. github provides distributed version control and zenodo provides automatic doi minting integrated with github releases. all research data and model outputs are additionally preserved in tu wien institutional repositories (dbrepo and tuwrd). P2 (NO2 Air Quality DBRepo Database) will be stored on DBRepo. External storage will be used because github and zenodo are used alongside institutional storage (dbrepo and tuwrd) for version control, collaboration, and software citation purposes. github provides distributed version control and zenodo provides automatic doi minting integrated with github releases. all research data and model outputs are additionally preserved in tu wien institutional repositories (dbrepo and tuwrd). **Data security and protection of sensitive data:** We pay strict attention to compliance with the relevant institutional and national data protection policies. At this stage, it is not foreseen to process any sensitive data in the project. If this changes, advice will be sought from the data protection specialist at TU Wien, and the DMP will be updated. Access to data during research: <table><tr><th>dataset ID</th><th>selected project members</th><th>all other project members</th><th>the public</th></tr><tr><td>P1</td><td>writing</td><td>reading only</td><td>reading only</td></tr><tr><td>P2</td><td>writing</td><td>reading only</td><td>reading only</td></tr><tr><td>R1</td><td>writing</td><td>reading only</td><td>reading only</td></tr></table>
IV.2 Data sharing and long-term preservation	**Data publication and access conditions:** As far as possible, obtained datasets will be published in repositories. Details on access conditions, reuse licenses, reasons for restrictions, etc. are collected in the table below. <table><tr><th>dataset ID</th><th>access conditions</th><th>estimated publication date</th><th>location for publication (repository)</th><th>PID</th><th>license</th></tr><tr><td>P1</td><td>Open</td><td>2026-03-30</td><td>TU Wien Research Data, Zenodo</td><td>DOI</td><td>CC-BY-4.0</td></tr></table>

P2	Open	2026-03-30	TU Wien Research Data	DOI	CC-BY-4.0

Repository description:

TU Wien Research Data is an institutional repository of TU Wien to enable storing, sharing and publishing of digital objects, in particular research data. It facilitates the funders' requirements for open access to research data and the FAIR principles by making research output findable, accessible, interoperable, and reusable. A DOI is assigned to each dataset published in TU Wien Research Data. This service is developed by the TU Wien Center for Research Data Management and hosted by TU.it. https://researchdata.tuwien.ac.at/

ZENODO builds and operates a simple and innovative service that enables researchers, scientists, EU projects and institutions to share and showcase multidisciplinary research results (data and publications) that are not part of the existing institutional or subject-based repositories of the research communities. ZENODO enables researchers, scientists, EU projects and institutions to: easily share the long tail of small research results in a wide variety of formats including text, spreadsheets, audio, video, and images across all fields of science. display their research results and get credited by making the research results citable and integrate them into existing reporting lines to funding agencies like the European Commission. easily access and reuse shared research results. https://zenodo.org/

Methods or software needed to access and use data: To access the raw data, users need Python with pandas and pyarrow libraries to read Parquet files. The DBRepo database can be accessed directly via the DBRepo REST API or web interface without any special software. The trained model requires Python with scikit-learn and joblib libraries to load and run predictions. All dependencies are documented in requirements.txt and the CodeMeta file. The GitHub repository, Zenodo archive, and TUWRD deposits are publicly accessible via standard web browsers without authentication.

Long-term preservation and deletion of data:

dataset ID	location for long-term storage	minimum retention period (≥ 10 years)	foreseeable research uses and/or users
P1	TU Wien Research Data, Zenodo	10 years	The primary target audience is researchers and data scientists working in environmental monitoring, air quality analysis, and FAIR data science. The EEA NO2 measurements can be reused for time series analysis, pollution trend studies, or training other machine learning models for air quality prediction across Austrian or European monitoring stations. The trained classifier and evaluation outputs can serve as a baseline for comparison in future NO2 classification experiments. The normalised DBRepo database provides a FAIR-compliant, query-ready version of the EEA data that can be reused without requiring raw Parquet file processing. Students and educators in data stewardship courses may also reuse the experiment as a reference implementation of FAIR principles.
P2	TU Wien Research Data	10 years	

V Legal and Ethical Aspects	
V.1 Legal aspects	**Personal data:** At this stage, it is not foreseen to process any personal data in the project. If this changes, advice will be sought from the data protection specialist at TU Wien, and the DMP will be updated. **Intellectual property rights and rights of use:** Legal restrictions on how data is processed and shared are specified in the EEA CC BY 4.0 licence terms available at https://www.eea.europa.eu/en/datahub/datahubitem-view/778ef9f5-6293-4846-badd-56a29c70880d and documented in docs/eea-licence-review.md in the project GitHub repository.. The restrictions relate to datasets R1 (EEA NO2 Air Quality Measurements Austria 2024). The legal restrictions are based on the following: The EEA NO2 input data is owned by the European Environment Agency and reused under CC BY 4.0. The project group (Muhammed Tayyab Sajjad, Sina Sadeghi, Glent Rembeci, Somayeh Zeraati) holds rights over the generated outputs including the trained model, predictions, and evaluation results, released under CC BY 4.0. The software and code is released under the MIT Licence. All datasets are publicly accessible via GitHub, Zenodo, DBRepo, and the TU Wien Research Data Repository. The following individual(s) hold rights and control access to the project data: EEA NO2 Air Quality Measurements Austria 2024: The European Environment Agency holds the rights as the original data publisher. The data is reused under CC BY 4.0 licence. The project group has the right to reuse, normalise, and redistribute the data with proper attribution. NO2 Air Quality Classifier and Evaluation Outputs: The project group (Muhammed Tayyab Sajjad, Sina Sadeghi, Glent Rembeci, Somayeh Zeraati) holds the rights and releases the outputs under CC BY 4.0. NO2 Air Quality DBRepo Database: The project group holds the rights to the normalised database representation. The underlying data remains under EEA CC BY 4.0 licence. All datasets are publicly accessible with no access restrictions.
V.2 Ethical aspects	**Ethical issues:** No particular ethical issue is foreseen with the data to be used or produced by the project. This section will be updated if issues arise.